

Architectures for VCC 2026

Fifteen ideas derived by deleting assumptions, with kill criteria

22 August 2026

How this document was made

Part I — The ideas

- Idea 1 — Programs, not vectors
- Idea 2 — Generation and cancellation as separate objects
- Idea 3 — Penetrance: a perturbation acts on a fraction of cells
- Idea 4 — Test-aware emission: invert the scorer
- Idea 5 — Transportability, not similarity
- Idea 6 — Context-gated graphs: the prior is not static
- Idea 7 — Predict the table, then realise it
- Idea 8 — Uncertainty *is* the dispersion
- Idea 9 — Context-specific essentiality as the magnitude prior
- Idea 10 — Predict n_{real} as a first-class task
- Idea 11 — Decision-theoretic posture per perturbation
- Idea 12 — The foundation model where it actually works
- Idea 13 — Knockdown efficacy as a latent scale
- Idea 14 — The modality-gap model
- Idea 15 — Ensembling across the structure, not the seeds

Part II — Ranking, sequencing, and what I would not do

- The ranking
- Suggested sequencing
- What I would not do, and why
- What I am least confident about

Appendix — The metric geometry, for reference

How this document was made

The method is deliberate and repeatable. For each established approach in the field I asked three questions:

1. **What is the load-bearing assumption?** Not the stated contribution — the thing that, if false, breaks the method.
2. **What happens if I delete it?** Not weaken it, delete it. What object appears in its place?
3. **Is the replacement principled, causally interpretable, and testable in the harness we built?**

The third filter is what separates this from brainstorming. Every idea below carries a **kill criterion**: a specific, cheap experiment on the leave-one-line-out harness that would tell you to abandon it. An idea without a kill criterion is a hobby, not a research direction. This is the discipline your earlier proof-of-concept had and the reason it was valuable even though its headline result was negative.

Two constraints govern everything and are worth restating because they are unusual:

The scoring geometry is lopsided. Roughly two-thirds of achievable score sits in the three DE-table metrics (jac ~29%, nmae ~18%, fid ~18%). Four of the six metrics are computed by running a Wilcoxon test on *the 400 cells you emit*, not on a mean profile. Cell-level dispersion is therefore load-bearing, and a model that predicts a perfect mean and emits it 400 times scores badly. Any architecture that outputs only a mean is incomplete by construction.

The only channel to a target context is an unperturbed count matrix. No in-context anchor, no calibration set, no outcomes to fit against. Every context-specific quantity — how hard perturbations bite here, how many genes respond, how noisy the cells are — must be *predicted from basal state* rather than estimated from outcomes. This is not a harder

version of 2025; it is a different problem, and it privileges methods that can condition on an observational snapshot.

A note on “causally interpretable.” I am using a specific and demanding standard throughout, taken from the failure of the LLM-reasoning family: **interpretability is only meaningful when the interpretable object is load-bearing — when a wrong mechanism necessarily produces a wrong prediction.** An architecture that emits an explanation alongside a prediction has produced decoration. An architecture where the prediction must pass *through* the mechanism has produced an explanation. Several ideas below are ranked highly specifically because they meet this bar, and I flag where an idea only appears to.

Part I — The ideas

Each idea states the assumption deleted, what emerges, why it is principled, how to test it, and what would kill it. Value estimates are my judgement, calibrated against the metric geometry, and are marked as such.

Idea 1 — Programs, not vectors

Deleted assumption: *a perturbation’s effect is a vector to be transported between contexts.*

Every transfer method — similarity-weighted aggregation, scGen’s latent arithmetic, kNN over reference lines — treats the effect of knocking down gene g as an object Δ_g that exists, gets measured in reference lines, and gets moved to the target with some correction. The vector is the primitive.

Delete that. Suppose instead the primitive is a small set of **response programs** — coordinated gene modules that perturbations *activate*, shared across contexts — and what varies between contexts is not the program’s content but its **activation weight**.

$$\Delta(g, c) \approx \sum_k w_k(g, c) \cdot P_k$$

where P_k are programs (fixed, learned once across all reference data) and $w_k(g, c)$ are activation weights depending on the perturbed gene and the context.

Why this is more than a reparameterisation. It splits the problem along exactly the fault line the field’s evidence identifies. The survey’s most robust finding on the hard fraction of perturbations is that *direction is recoverable and magnitude is not*. In this factorisation, programs carry direction and co-regulation structure (which genes move together, with what signs); weights carry magnitude. You can therefore be confident about P_k and aggressively shrink w_k — which is precisely the optimal policy under `nmae`’s asymmetric floor. In the vector formulation, direction and magnitude are entangled in every coordinate and you cannot shrink one without damaging the other.

It also explains, rather than merely exploits, why the training-mean baseline is so strong: the shared stress/quiescence response is a program with near-constant weight across perturbations. Predicting the mean is predicting P_1 with an average weight. Your model then only has to beat the baseline on the *residual* programs, which is a much better-posed problem than beating it on 18,533 correlated coordinates.

Causal interpretability. Real if the programs are grounded in nameable biology (pathway annotations, known co-regulated modules) rather than free latent factors, and load-bearing in the strong sense: the prediction passes through the program decomposition, so assigning the wrong programs necessarily produces a wrong prediction. This is the “load-bearing mechanism” criterion satisfied in a cheap, non-exotic way. Note the difference from SAMS-VAE-style latent sparsity, where “this perturbation affects dimension 12” is not a biological statement — here dimension 12 has a name.

How to build it. Non-negative matrix factorisation (or sparse PCA, or a pathway-masked linear decomposition) on the stacked reference delta matrix across all lines and perturbations. Then a regressor from (gene identity features, basal-state features) to weights. Both halves are small and CPU-only.

Test: does the program model beat flat similarity-weighted transfer on held-out lines? Additionally: are programs stable across leave-one-dataset-out splits (compute program-to-program correlation between fits) — an unstable basis is a fitted artefact.

Kill criterion: if reconstruction of held-out-line deltas from ≤ 30 programs is no better than the rank-equivalent PCA of the training-mean, or if programs do not transfer across the dataset-out split, abandon. **Estimated value: high.** This is my top structural idea and it composes with almost everything else below.

Idea 2 — Generation and cancellation as separate objects

Deleted assumption: *a small observed response is a small response.*

Every model in the field represents non-response as a number near zero. The survey names this as one of five unsolved problems: inert and compensated perturbations are indistinguishable objects in every current architecture. Delete the assumption that the observed delta is the primitive, and model it as a difference:

$$\Delta_{\text{obs}} = \Delta_{\text{naive}} - C$$

where Δ_{naive} is the response that propagation alone would produce and C is a cancellation term from homeostatic feedback and redundancy. A gene that is inert has $\Delta_{\text{naive}} \approx 0$. A gene that is buffered has Δ_{naive} large and $C \approx \Delta_{\text{naive}}$. These become distinguishable objects.

The honest problem, stated up front. At steady state you observe only the difference. Δ_{naive} and C are **not identifiable** without constraints — this is the same identifiability wall that causal representation learning hits, and pretending otherwise would be exactly the “mechanism as decoration” failure this document is trying to avoid. The idea is only viable with a constraint that makes the split observable, and there is one available:

C must be a function of named, basal-expressed substitutes. Not a free-form learned residual — a term computed from the expression of specific paralogues and pathway-parallel genes in *this* context’s control profile. Under that restriction the decomposition has content, because C is forced to vary with something you can independently measure (whether the backup is switched on here) rather than absorbing whatever the fit needs.

Causal interpretability. Strong when the constraint holds, and load-bearing: get the buffering wrong and the prediction is wrong by construction. This is the closest thing in this document to the survey’s “architectures that represent non-response mechanistically” open problem.

Test: across held-out lines, does a model with the two-term structure beat a single-term model of equal capacity at predicting *which* perturbations produce near-zero response? Score specifically on the weak-effect subset, where `jac` and `fid` are most fragile.

Kill criterion: if the fitted C has no measurable dependence on substitute expression (i.e. it degenerates into an arbitrary residual with the same fit as an unconstrained model), the decomposition is decorative and should be deleted.

Estimated value: high if it survives, and it is the most likely of these ideas to be publishable independently of the competition. Also the most likely to fail its identifiability test — treat that as informative, not as wasted effort.

Idea 3 — Penetrance: a perturbation acts on a fraction of cells

Deleted assumption: *all 400 cells receiving a perturbation are drawn from one perturbed state.*

Replace with a mixture: a fraction π of cells enter a responder state, the remaining $1 - \pi$ remain effectively unperturbed.

$$\text{cells} \sim \pi \cdot D_{\text{responder}} + (1 - \pi) \cdot D_{\text{control}}$$

This is biologically real — incomplete knockdown, cell-cycle-dependent competence, and the well-documented observation that a perturbation may push 30% of cells into a distinct state while leaving the rest untouched — and it has an outsized mechanical consequence under the 2026 metrics.

Why it matters more here than anywhere else. Your DE table is computed by running Wilcoxon on the cells you emit. The number of genes you call significant, n_{pred} , is a function of your emitted *dispersion*, and both `fid` and `jac` are maximised at $n_{\text{pred}} \approx n_{\text{real}}$. A single interpretable scalar π per (perturbation, context) controls that dispersion directly: low π produces a bimodal, high-variance sample that under-calls; high π produces a tight sample that over-calls. You have converted a delicate noise-calibration problem into estimating one biologically meaningful number.

It also gives non-response a second, cleaner representation: $\pi \approx 0$ (nothing responded) is distinguishable from “small shift in every cell.” Note that this is a *different* mechanism from Idea 2’s buffering — penetrance is heterogeneity across cells, buffering is cancellation within a cell — and the two are complementary rather than competing.

Causal interpretability. Moderate and honest. π is a real, nameable quantity with an experimental interpretation. It

does not explain *why* penetrance is what it is, but it does not pretend to.

Test: fit π per perturbation on reference lines (mixture fit on real perturbed cells), check the distribution is non-degenerate, then predict π from basal state on held-out lines and measure whether emitted n_{pred} tracks n_{real} better than a fixed-dispersion decoder.

Kill criterion: if fitted π is close to 1 for nearly all perturbations in reference data, the mixture is unnecessary and a simpler dispersion model wins. **Estimated value: high, and unusually cheap — this sits in the counts-decoder slot, which is the highest score-impact-per-effort component you own and has almost no biological content.**

Idea 4 — Test-aware emission: invert the scorer

Deleted assumption: *predict expression accurately and the DE metrics take care of themselves.*

They do not. There is a lossy channel between your belief and your score: you emit counts, the scorer runs rank-sum tests against real controls, and *that* produces the table four of six metrics read. A well-calibrated belief passed through a badly-specified emission step arrives as a bad DE table.

Delete the assumption and treat emission as a **constrained sampling problem**: given a believed significance set, per-gene signs, magnitudes, and confidences, construct 400 cells whose Wilcoxon-vs-real-controls reproduces that belief as closely as possible, subject to the contract (integer counts, sparsity, density cap, library-size cap).

Is this gaming? No, and the distinction is worth being precise about because it determines whether the idea is legitimate. Gaming is manufacturing score without carrying perturbation-specific information — spiking the other 299 target genes, raising every below-cutoff gene identically. Every such route has been closed, and deliberately. This is the opposite: it is *avoiding the loss* of information you genuinely have to a mis-specified noise model. You are reporting your honest belief through the channel the scorer actually reads. A submission with no signal gains nothing from a better emission step.

Causal interpretability. None, and none claimed. This is a measurement-model component, and it should be kept architecturally separate from the biological components precisely so that neither contaminates the other.

Test: take *real* reference perturbation data, compute its true DE table, then discard the cells and try to regenerate 400 cells from the table alone. Score the regenerated cells. The gap to the real cells' score is your emission loss — and it is an upper bound on what this component can win you.

Kill criterion: if that regeneration gap is small (say under 5% of achievable score), emission is already near-lossless and effort belongs elsewhere. I doubt this, but it is measurable in an afternoon and you should measure before building.

Estimated value: high, and it is the single most separable component in the pipeline — it can be developed and validated against the public scorer while every biological question is still open. Build it early for that reason alone.

Idea 5 — Transportability, not similarity

Deleted assumption: *reference lines should be weighted by global transcriptome similarity to the target.*

This is the core of the statistical transfer baseline and it embeds a strong claim: that a line similar overall is a good donor for *every* gene. That claim is false in an interesting way. Two lines can be globally similar yet differ precisely in the pathway that gene g acts through; two globally dissimilar lines may share g 's regulatory neighbourhood entirely.

Delete global similarity and make the weight **gene-specific**: to predict the effect of knocking down g in context c , weight reference line c' by similarity restricted to g 's functional neighbourhood — its pathway members, its known interactors, the genes it regulates.

$$w(c', g) = \text{sim}(x_{c'}[N(g)], x_c[N(g)])$$

Why this is principled rather than a heuristic. It is a statement about *transportability* in the causal-inference sense: an effect transfers between populations when the mechanism generating it is invariant across them, not when the populations are marginally similar. Restricting the similarity computation to the mechanism's neighbourhood is a direct, if approximate, operationalisation of that. This is a better justification than "graphs are biologically motivated," which is the justification GEARS-style models rely on and which the benchmarks did not reward.

Causal interpretability. Moderate-to-good, and it is legible: for any prediction you can say which reference lines were used and which genes' agreement justified it. A wrong neighbourhood produces a wrong donor set produces a wrong prediction — load-bearing.

Test: direct A/B against global similarity weighting on the harness, held out per line and per dataset. This is a small change to a component you have already built.

Kill criterion: if gene-specific weighting does not beat global weighting on `pds` and `jac` across held-out lines, drop it — and that result would itself be informative, since it suggests effect transfer is governed by global cell state rather than local pathway state.

Estimated value: high value per unit effort. Of everything here, this has the best ratio of (potential gain) to (hours to implement), because the surrounding machinery already exists.

Idea 6 — Context-gated graphs: the prior is not static

Deleted assumption: *the knowledge graph is the same in every cell type.*

This is the flaw the survey names directly: GO annotations and STRING edges do not vary by cell type, but regulatory relationships do. Every graph-propagation model inherits it.

Delete it using the one thing 2026 gives you: **mask edges by basal expression.** A regulatory edge $u \rightarrow v$ cannot act in a context where u is not expressed. So gate every edge by the expression of its endpoints in the target line's control profile, then propagate over the gated graph.

Why it is attractive. It converts a static prior into a context-specific one *with zero new parameters and no extra data*, and the justification is close to mechanical rather than statistical — an unexpressed regulator cannot regulate. It also directly addresses the “closed world, context-free graph” critique without needing the proprietary graphs that TxPert's strongest results depend on and that no academic group can reproduce.

A caution that should temper enthusiasm. A model propagating along arbitrarily-oriented edges is performing similarity smoothing, not causal propagation — this is true of several public graph types, by their own authors' admission. Gating by expression makes the smoothing context-aware; it does not make it causal. Be honest in how you describe it, and prefer graph sources with genuine directionality (regulatory networks) over undirected association graphs where a causal reading is claimed.

Test: ablate gated vs ungated propagation, same architecture, multiple seeds, on held-out lines.

Kill criterion: no measurable difference between gated and ungated. **Estimated value: moderate, but very cheap, and it is the most defensible single upgrade to the graph family in this setting.**

Idea 7 — Predict the table, then realise it

Deleted assumption: *the model's output is an expression matrix.*

Four of six metrics read the DE table. The expression metric is worth about a ninth of the score and is capped, so accuracy past replicate level earns nothing. Yet nearly every architecture in the field is built to emit expression, from which a table is incidentally derived.

Invert it. Make the **DE table the primary predicted object**, decomposed into three well-posed sub-problems:

1. *Which genes respond* — binary classification per gene (drives `jac`).
2. *In which direction* — sign, with a confidence ordering (drives `fid` and `reach`).
3. *By how much* — magnitude, shrunk (drives `nmae`).

Then a realisation step (Idea 4) synthesises cells consistent with the table.

Why this is a real gain and not bookkeeping. Each sub-problem gets its own supervision signal, its own baseline, and its own error analysis. Sub-problem 2 is where the field's evidence says models genuinely add value — deep models recover *which genes respond and in which direction* on the resistant fraction, but not magnitudes. Structuring the output this way lets you spend capacity where it demonstrably works and apply conservative shrinkage where it demonstrably does not, rather than asking one loss to arbitrate.

The `reach` metric is worth special attention here: it needs only a sensible per-gene confidence *ordering*, has a near-zero baseline and a negligible floor, and is close to free. A table-first architecture produces that ordering natively; an expression-first architecture has to reverse-engineer it.

Causal interpretability. Neutral in itself, but it makes room for the mechanistic components — a GRN-propagation module feeding sub-problem 2 is a natural fit, and its documented weakness (direction yes, magnitude no) is precisely aligned with the sub-problem it would serve.

Test: build the three heads on top of the existing transfer estimator; compare against deriving the table from a predicted expression matrix.

Kill criterion: if table-first and expression-first score the same, prefer expression-first for simplicity. **Estimated value: high — this is the organising principle I would use for the pipeline as a whole rather than a single component.**

Idea 8 — Uncertainty is the dispersion

Deleted assumption: *predictive uncertainty is a diagnostic to report alongside the prediction.*

In this competition uncertainty is not a side output — it is a *submitted quantity*. The spread of the 400 cells you emit determines how many genes the scorer calls significant, which determines `jac` and `fid`. So your posterior width and your emitted dispersion should be the same object.

Delete the separation. Predict a posterior over each gene's effect; let the emitted cells be samples that carry that posterior's width. A perturbation you are confident about produces a tight sample and many significant calls; one you are unsure about produces a diffuse sample and fewer.

Why this unifies several problems at once. It makes shrinkage automatic and principled rather than a fitted global constant — an empirical-Bayes or James-Stein posterior shrinks unreliable estimates toward zero *by construction*, which is exactly the policy `nmae`'s asymmetric floor demands (predicting zero fold change everywhere scores exactly the baseline; being wrong in magnitude is punished with a floor at -6). And it couples the shrinkage factor to the calling volume, which the briefing correctly notes are *not independent knobs*. Treating them as one object is more honest than tuning them separately.

The trap to avoid. Under-calling is not conservative here. A genuine half of the reference scores -0.15 to -0.34 on `fid` because it calls too few genes. So the posterior must be calibrated, not merely wide: diffuse-everything is a losing policy, and “be uncertain” is not automatically safe. Calibration must be validated on held-out lines, not assumed.

Test: compare (a) point prediction + fitted global shrinkage against (b) posterior-driven emission, on overall score and on the `npred / nreal` ratio.

Kill criterion: if calibrated uncertainty does not improve the call-volume match over a single global dispersion parameter, the extra machinery is not paying for itself. **Estimated value: high, and it is the most elegant available answer to the shrinkage problem — but it must be validated, not assumed.**

Idea 9 — Context-specific essentiality as the magnitude prior

Deleted assumption: *effect magnitude must be learned from perturbation data.*

There is an adjacent, better-studied prediction problem with far more data: **is gene g essential in cell context c?** Large essentiality screens across hundreds of cell lines exist, and predicting context-specific essentiality from basal expression is a comparatively mature task.

Delete the assumption that magnitude comes only from Perturb-seq, and import essentiality as the prior on effect size: essential-here genes produce large responses, non-essential-here genes produce small ones.

Why this is a strong play for a compute-limited individual. It routes around the field's binding constraint. You have on the order of ten cell lines with full perturbation outcomes — that is the scarce resource, and it is why every model in this space is data-starved along the context axis. Essentiality data spans *hundreds* of lines. If magnitude is substantially predicted by context-specific essentiality, you have imported a much larger effective training set through a side channel, using exactly the input 2026 gives you (basal expression).

Causal interpretability. Moderate. “This knockdown will produce a large response because this gene is essential in cells with this expression profile” is a real, checkable claim, though essentiality is a phenotype rather than a mechanism.

Test: on reference lines, regress observed response magnitude on predicted essentiality; then check the predictor transfers to held-out lines.

Kill criterion: weak correlation between essentiality and transcriptomic response magnitude on reference data. Worth

checking first, since it is a one-afternoon analysis and it gates the whole idea.

Leakage warning: build the essentiality predictor from screens that exclude the held-out line, or you will measure your own answer. This is easy to get wrong and it invalidates the harness silently.

Estimated value: high, and strategically distinctive — most competitors will be fighting over the same ten Perturb-seq datasets.

Idea 10 — Predict n_{real} as a first-class task

Deleted assumption: *how many genes respond is a by-product of predicting which genes respond.*

It need not be. “How many genes respond to knocking down g in an unseen context?” is a scalar regression problem with direct score consequences on two metrics, and it is *easier* than the vector problem — you can be right about the count while uncertain about the membership, and the count alone fixes your calling threshold.

The briefing flags this as, to its author’s knowledge, not posed as a task in its own right. You now have a harness that can measure it on real data across held-out lines. That combination — an unclaimed, well-posed question and a working evaluator — is the cleanest publishable opportunity in this document, and it is independent of whether you win anything.

How it plugs in. An accurate n_{real} estimate sets the threshold at which you cut your ranked gene list, putting you at the $n_{\text{pred}} \approx n_{\text{real}}$ optimum that both `fid` and `jac` want. It is also the natural output of Ideas 3 and 9 (penetrance and essentiality both predict response extent), so it is a convergence point rather than a separate limb.

Test: predict n_{real} from basal state + gene features on held-out lines; compare against the training-mean n_{real} baseline. Report correlation and calibration.

Kill criterion: if you cannot beat “predict the average number of responding genes,” the sub-task is not learnable from basal state and the calling threshold should just be set to that average. **Estimated value: high, with the highest ratio of research interest to implementation cost here.**

Idea 11 — Decision-theoretic posture per perturbation

Deleted assumption: *the model should apply the same policy to every perturbation.*

The metric floors are wildly asymmetric. `nmae` is floored at -6 and a raw value of 2.0 already costs roughly -1.7 ; `fid` has the deepest unclamped floor and the narrowest span, making it volatile; `reach` and `jac` have negligible floors and cannot meaningfully hurt you; the expression metric is clamped at 0 and cannot be gamed.

So the *optimal action depends on your confidence*, and confidence varies enormously across perturbations. Delete the uniform-policy assumption and choose a posture per perturbation: where you have validated magnitude skill, predict magnitude and earn above the `nmae` baseline; where you do not, shrink to zero and take the safe 0 — while still calling genes, because silence is punished on `fid`.

Why this is legitimate optimisation rather than metric-hacking. It requires *knowing which perturbations you are reliable on*, which is a genuine modelling capability (and closely related to Idea 8). You cannot execute this policy without real self-knowledge; there is no free lunch in it. Contrast with the closed loopholes, which required no perturbation-specific information at all.

Test: split held-out perturbations by predicted confidence, apply differentiated shrinkage, compare against uniform shrinkage.

Kill criterion: if your confidence estimate does not correlate with actual per-perturbation error on held-out lines, you have no basis for differentiating and should apply the uniform policy. **Estimated value: moderate-to-high, and it is pure upside once Idea 8 exists — a thin decision layer over machinery you already have.**

Idea 12 — The foundation model where it actually works

Deleted assumption: *foundation models are unsuitable here because pretraining is correlational and prediction requires causal structure.*

The critique is correct as stated and I am not disputing it: masked-expression pretraining teaches which genes co-vary in unperturbed cells; perturbation prediction needs what happens when you intervene; correlation identifies causation

only under assumptions transcriptomic data do not supply, and more parameters do not create a distinction the target variable does not encode.

But notice *which sub-problem* you would use it for. VCC 2026's distinctive demand is: **represent an unseen cell context from its unperturbed transcriptome alone**. That is an observational task. It is exactly what pretraining on hundreds of millions of unlabelled cells is good at, and the causal part of the problem is shared equally by every method family — the foundation model is not being asked to supply it.

So: use a pretrained encoder *only* as the context encoder (Idea 5's similarity space, Idea 9's essentiality features, Idea 10's n_{real} predictor all consume a context representation), and keep every causal claim in components that are explicitly structured. This also matches the empirical record — the winning 2025 entry was a foundation-model variant *augmented with* protein embeddings and curated external perturbation data, i.e. a hybrid, and Arc's own retrospective concluded hybrids beat pure end-to-end learning.

Test: the retrieval test. Given a held-out line's controls, does the encoder rank public reference lines by true similarity better than pseudobulk correlation does? If it cannot retrieve, it cannot condition.

Kill criterion: no improvement over simple pseudobulk features on retrieval or on downstream held-out score. Given the benchmark record where multiple foundation models failed to beat linear baselines, treat this as a genuine possibility rather than a formality.

Estimated value: moderate. Worth doing, worth not believing in advance. Note the compute asymmetry: this is the only idea here with a meaningful GPU cost, which is another reason to gate it behind a cheap retrieval test.

Idea 13 — Knockdown efficacy as a latent scale

Deleted assumption: *a knockdown is a fixed intervention with the same strength everywhere.*

CRISPRi knockdown is partial and variable — typically 60-95%, varying by guide. Arc selected the 2026 panel for greater than 80% on-target knockdown, which bounds but does not eliminate the variation. And the target gene's own transcript drop is *directly observable in reference data* even though it is excluded from scoring.

Delete the fixed-intervention assumption and treat achieved knockdown as a latent scale on the response: predicted magnitude should scale with achieved suppression. In reference data you can measure it; in the target context you cannot, but the >80% selection gives you a prior.

Why it is worth the trouble. It cleans your *training signal*. Reference effect sizes are contaminated by variable knockdown efficacy, and normalising them by observed on-target drop removes a nuisance factor before you ever try to transfer. Failing to do this is the most common source of contamination in claims about which genes do nothing.

Causal interpretability. Good — this is dose-response, the most classically interpretable structure in pharmacology.

Test: on reference lines, regress response magnitude on observed target-gene drop; check whether efficacy-normalised effect vectors transfer better across lines than raw ones.

Kill criterion: no relationship between achieved knockdown and response magnitude in reference data. **Estimated value: moderate, mostly as data hygiene rather than as an architecture — but hygiene that improves every downstream component.**

Idea 14 — The modality-gap model

Deleted assumption: *reference data from different assays can be pooled once harmonised.*

Your corpus mixes CRISPRi at different chemistries, and the wider literature adds CRISPRa and knockouts. These are not interchangeable: a knockout produces mutant mRNA whose degradation triggers upregulation of sequence-similar genes (transcriptional adaptation), a compensation CRISPRi does not invoke — which is *why* knockouts often show milder phenotypes than knockdowns of the same gene. And CRISPRa is the opposite perturbation direction entirely.

Delete the poolability assumption and model the modality gap explicitly as a learned correction term, fitted where the same gene is measured under two modalities.

Why it matters for you specifically. It determines how much data you actually have. If the gap is small and correctable, a large amount of otherwise-unusable public data becomes admissible reference material — and data along the context axis is your binding constraint. If it is large, you should restrict to matched-modality sources and stop pretending the rest helps.

Test: find genes measured in both CRISPRi and another modality in overlapping lines; quantify the gap; test whether a fitted correction makes cross-modality references useful on held-out lines.

Kill criterion: if corrected cross-modality data does not beat matched-modality-only training, exclude it and say so.

Estimated value: moderate, but it is *decision-relevant* — it tells you the size of your usable corpus, which affects every other idea.

Idea 15 — Ensembling across the structure, not the seeds

Deleted assumption: *ensembling means averaging several runs of the same model.*

The metrics reward different capabilities, and the evidence says different families have complementary strengths: mechanistic GRN propagation is good at direction and cell-fate shifts but poor at magnitude; statistical transfer is good at magnitude and calibration; graph propagation is good at relating genes. Meanwhile `fid` is insensitive to the order of calls while `reach` reads only the order — so a submission strong on one can be weak on the other.

Delete seed-ensembling and ensemble *by sub-problem*: let the GRN/graph channel produce the direction and confidence ordering, let the statistical channel produce magnitude and calibration, let the mixture model produce dispersion. Each component is assigned the sub-problem the evidence says it is good at.

Why this is the shape the 2026 metrics reward and 2025's did not. In 2025, magnitude-style errors dominated the headline metric and a direction-only module had nowhere to contribute. In 2026 three of the four DE metrics read direction and membership only, carrying the majority of achievable score, while the magnitude metric can be safely shrunk toward its baseline. A direction-strong, magnitude-weak component is now a genuine asset rather than a liability.

Causal interpretability. Good, and this is the payoff: the mechanistic component is confined to the sub-problem where its mechanism is load-bearing, instead of being asked to carry quantitative prediction it cannot support and then being blamed for failing.

Test: per-sub-problem ablation — swap each channel for a trivial alternative and measure the drop.

Kill criterion: if the mechanistic direction channel does not beat a statistical direction channel on `fid` and `reach`, drop it and simplify. **Estimated value: high as an integration strategy, and it is how the individual ideas above become one system rather than fifteen experiments.**

Part II — Ranking, sequencing, and what I would not do

The ranking

Judgement, not measurement. “Effort” assumes the harness exists.

#	Idea	Score impact	Effort	Causal?	Verdict
4	Test-aware emission	Very high	Low	No	Build first
3	Penetrance mixture	High	Low	Partial	Build first
5	Transportability weighting	High	Very low	Yes	Build first
7	Table-first output	High	Medium	Neutral	Organising principle
1	Programs not vectors	High	Medium	Yes	Core structural bet
10	Predict n_{real}	High	Low	Partial	Research opening
8	Uncertainty as dispersion	High	Medium	No	Unifies shrinkage
9	Essentiality prior	High	Medium	Partial	Strategically distinctive
15	Sub-problem ensembling	High	Medium	Yes	Integration strategy
2	Generation/cancellation	High if it survives	High	Strongly	Research bet
11	Decision posture	Moderate-high	Low	No	Cheap upside
13	Efficacy normalisation	Moderate	Low	Yes	Data hygiene
6	Context-gated graphs	Moderate	Low	Partial	Best graph upgrade
14	Modality gap	Moderate	Medium	No	Decision-relevant
12	Foundation encoder	Moderate	High (GPU)	No	Gate behind retrieval test

Suggested sequencing

Week 1 — measurement components. Ideas 4 and 3. Both are nearly free of biology, both have the largest score impact per hour, and both are validated against the public scorer with matched-chemistry data. A bug here silently destroys four of six metrics no matter how good the biology is, which is the strongest possible argument for doing it first.

Week 2 — cheap wins on existing machinery. Ideas 5 and 13, plus the one-afternoon checks that gate Ideas 9 and 14. These are small changes to components you have, and their results *redirect* later effort.

Weeks 3-4 — the structural bet. Ideas 7 and 1 together: restructure the output as a DE table, and factor effects into programs and weights. This is the point where the architecture stops being a baseline with patches.

Weeks 5+ — the research bets. Ideas 10 and 2, plus 8 and 11 as the decision layer, and 15 to integrate. Idea 12 only if a retrieval test justifies the GPU time.

The ordering is deliberate: **measurement, then transfer, then structure, then research.** Each stage's results change what the next stage should be, which is the property you want when you have three months and no in-context feedback.

What I would not do, and why

Do not build a large end-to-end trainable model. Zero in-challenge training data, ten-ish reference lines, and a benchmark record where multiple foundation models and purpose-built deep models failed to beat linear baselines. Capacity is not the missing ingredient; constraint is.

Do not resurrect the KG-GNN as the centrepiece. Your own proof-of-concept found the prior's advantage collapsed under a capacity-matched comparison, TxPert's strongest results lean on proprietary graphs, and its edge directions are partly arbitrary. Ideas 6 and 15 are the defensible remnants: gate the graph by context, and confine it to the direction sub-problem where its contribution is load-bearing.

Do not chase the expression metric. It is worth about a ninth of the score, clamped at 1, and cannot be gamed. It comes along with a good model.

Do not tune hyperparameters on the leaderboard. Validation and final scores are formally incomparable — different lines, different panels, and perturbation discrimination is a rank within its own panel. Prefer leave-one-line-out

cross-validation on public data. Your harness is worth more than either daily submission.

Do not model genetic interactions. No combinations in 2026. This is the field’s worst-performing area and this year it is a free pass; spending effort here is spending on the one problem you are not being asked.

What I am least confident about

Idea 2’s identifiability is the shakiest claim in this document, and I have tried to flag it rather than smooth it over. The constraint I proposed — forcing the cancellation term to depend on named substitute expression — may not be enough to separate the two channels from steady-state data, and it may fail in a way that is hard to detect (the model fits well, the decomposition is meaningless). Treat the identifiability check as a genuine experiment with a real chance of a negative result.

Idea 9’s premise, that context-specific essentiality predicts transcriptomic response magnitude, is plausible and I believe it, but I have not verified the strength of that relationship. It is a one-afternoon check and it gates a large idea. Do the check before building.

Idea 1’s programs may turn out to be dominated by the shared stress response, with the residual programs too weak or too unstable to transfer. That would be a real negative result and worth knowing early — it would say the perturbation-specific signal is thinner than the factorisation assumes.

More generally: several of these ideas draw on 2026 preprints and derived arithmetic rather than settled results, and the field’s own benchmarking literature is actively contested — metric choice has repeatedly flipped conclusions. Nothing here should be believed over your harness.

Appendix — The metric geometry, for reference

Anchors are per-context; scaled score is $(u - b) / (r - b)$ where b is the context-mean baseline and r is a real replicate experiment.

Metric	Baseline b	Replicate r	Ceiling	Floor	Share
de_wilcoxon_sig_jaccard	0.021-0.037	0.375-0.423	2.48-2.85	-0.05 to -0.11	~29%
de_wilcoxon_lfc_nmae	1.0009-1.0017	0.369-0.431	1.58-1.75	-6	~18%
de_wilcoxon_direction_fidelity	0.505-0.522	0.795-0.832	1.51-1.72	-1.55 to -1.85	~18%
pds_cosine	0.500	0.927-0.984	1.03-1.17	-1.03 to -1.17	~12%
de_wilcoxon_direction_reach	0.047-0.097	0.958-0.978	1.02-1.05	-0.12	~11%
expr_mse_unbiased_capped_norm	0.986-0.992	0.028-0.045	1.000	0 (clamped)	~11%

A perfect submission scores roughly 1.44-1.59, not 1.0. Negative scores are normal. Catastrophic downside is concentrated in `nmae` and `fid`; `reach`, `jac` and the expression error cannot meaningfully hurt you.

Three consequences that should shape architecture:

- 1. **Two-thirds of achievable score is in the DE table**, so the table deserves to be the modelled object (Idea 7).
- 2. **`nmae`’s asymmetry makes shrinkage the default policy** — predicting zero fold change scores exactly the baseline, while being wrong scores unboundedly worse until raw 4.4 (Ideas 8, 11).
- 3. **Dispersion of emitted cells drives four of six metrics**, so the counts decoder is a first-class model component, not plumbing (Ideas 3, 4).

Metric values are taken from the technical briefing’s reading of the vcc2026 specification; verify against the published spec before relying on them quantitatively.